Safe Variable-Force Control for Robotic Ultrasound Scanning via ℓ_2-Regularized Fuzzy Broad Offline Reinforcement Learning

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***Abstract*—Robotic ultrasound scanning (RUSS) is a promising technology for standardizing examinations, improving repeatability, and alleviating the workload of sonographers. A key prerequisite for its safe clinical use is reliable control of the contact force between the probe and the patient's skin. In practice, this force should vary along the scan: it must be reduced over symptomatic or painful regions to avoid discomfort and potential tissue damage, while remaining sufficiently high elsewhere to ensure good acoustic coupling and image quality. Patient-specific soft-tissue properties and unstructured examination environments make it difficult to tune conventional (adaptive) impedance controllers, often leading to force overshoot or poor tracking performance. To address this challenge, we propose an ℓ_2-regularized fuzzy broad offline reinforcement learning framework (ℓ_2-FBORL) that learns variable force control policies from limited, suboptimal demonstrations generated by stabilizing controllers. Building on ℓ_2-FBORL, we formulate variable force tracking in RUSS as a Markov decision process and develop a Fuzzy Learning-based Adaptive Impedance Control (FLAIC) scheme that adaptively adjusts the impedance adaptation rate to suppress overshoot while maintaining fast convergence. The proposed approach is evaluated in high-fidelity simulation, phantom experiments, and in vivo upper-limb venous and abdominal scanning under respiratory motion. Across all scenarios, FLAIC reduces force overshoot and improves tracking accuracy compared with conventional impedance and adaptive impedance controllers, without patient-specific retuning, indicating its potential to enhance the safety, robustness, and standardization of robotic ultrasound procedures.**

***Index Terms*—Robotic ultrasound scanning, force tracking, offline reinforcement learning, fuzzy critic design.**

I. Introduction

ROBOTIC ultrasound scanning (RUSS) has emerged as a promising approach to standardizing ultrasound examinations, improving repeatability, and alleviating the workload of sonographers [1], [2]. Representative RUSS applications have been explored across multiple anatomies, including thyroid [3], vascular [4], carotid [5], lungs [6], heart [7], abdominal organ [8] and so on. Unlike conventional manual scanning, where the clinician implicitly adjusts both probe pose and contact force, a RUSS system must explicitly regulate the interaction between the ultrasound (US) probe and the patient's skin to balance image quality, patient comfort, and safety [2]. In many clinically relevant scenarios, a constant target force is not adequate. Instead, the desired contact force should vary along the scan: it needs to be reduced over symptomatic or painful regions to avoid additional discomfort or potential tissue damage, while remaining sufficiently high in non-symptomatic regions to ensure good acoustic coupling and maintain diagnostic image quality, as illustrated in Fig. 1. Designing controllers that can achieve such spatially varying, clinically appropriate force profiles in the presence of patient-specific soft-tissue properties and unstructured examination environments motivates the variable force tracking problem studied in this work.

Despite these advantages, achieving reliable variable force tracking in RUSS remains challenging. First, the human–robot interaction environment is inherently unstructured [9], and the nonlinear, patient-specific properties of soft tissue can substantially degrade the performance of conventional force tracking control methods. Second, when the target contact force is adjusted, existing methods often exhibit large force overshoot [10], which raises safety concerns in patient-facing applications. Consequently, it is critical for clinically viable RUSS systems to minimize or avoid force overshoot during force transitions while preserving high tracking accuracy across different patients and anatomical sites.

Force tracking control in robotic systems is most commonly built upon impedance control (IC) [11]. To improve robustness against uncertainties in robot dynamics and envi-

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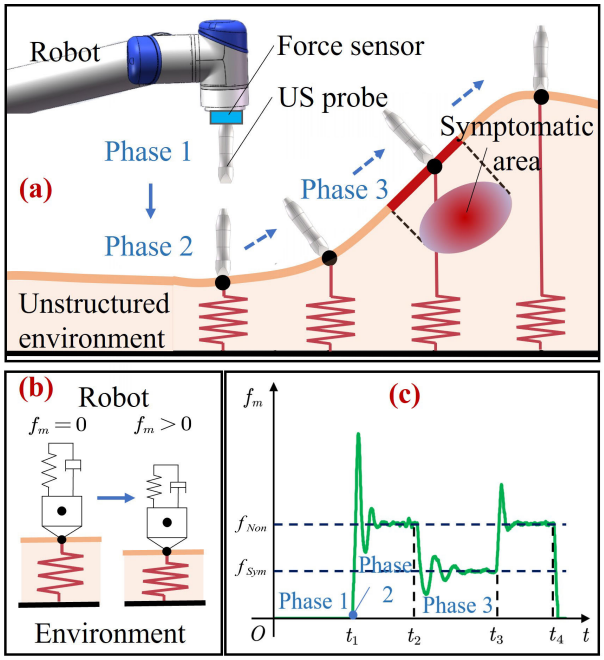


Fig. 1. Robotic variable force tracking in medical US scanning. (a) Task description of robotic variable force tracking. (b) Robot-environment contact models. (c) Force curve example in robotic US scanning.

ronment stiffness/position, a variety of IC-based schemes have been proposed [10], [12], [13]. Among them, adaptive impedance control (AIC) [10], [12], [13] adjusts impedance parameters online and has shown clear advantages over conventional IC. However, the behavior of AIC is highly sensitive to parameter tuning; inappropriate settings can lead to substantial overshoot and degraded tracking accuracy [10], which, in a clinical context, may cause patient discomfort or even safety incidents.

Reinforcement learning (RL) has been combined with IC to realize variable impedance control, e.g., via PI^2-based learning of joint-space impedance in simulation [14], analyses of different action spaces with sim-to-real validation [15], and RL-based IC schemes for unknown terrains using randomized contact environments in CoppeliaSim to enhance force tracking robustness [16]. For RUSS, however, the unstructured and patient-specific nature of human–robot interaction makes accurate robot–tissue modeling and safe online exploration difficult, limiting the practicality of these online RL-based approaches. Offline reinforcement learning (ORL) is therefore more suitable for medical scenarios, as it learns policies from fixed datasets, such as expert demonstrations and trajectories from stabilizing controllers [17], [18]. Although ORL has been applied to various robotic control tasks [19], [20], [21], [22], it has rarely been explored for robotic force tracking, which places greater emphasis on designing powerful critics to obtain reliable value estimates and robust policies [23], [24].

Recently, the fuzzy broad learning system (FBLS) [25] has been proposed on the basis of the broad learning system (BLS) [26] as an ensemble framework for modeling and controlling highly nonlinear and complex systems [27], [28]. FBLS combines fuzzy learning systems with random vector functional link (RVFL) neural networks in a wide, non-deep architecture, and has been successfully applied in several real-world control and prediction problems [29], [30], [31]. These properties make FBLS a promising candidate for critic approximation in data-limited medical robotic scenarios.

Drawing on the strengths of both ORL and FBLS, we develop an ℓ_2-regularized fuzzy broad offline reinforcement learning method, termed ℓ_2-FBORL, to address the variable force tracking problem in RUSS. The main contributions of this work are threefold:

1) **ℓ_2-FBORL for offline policy learning:** We propose an ℓ_2-regularized FBLS-based ORL algorithm that enables efficient policy learning from limited, potentially suboptimal offline data. To the best of our knowledge, this is the first ORL method that employs FBLS as the critic for robotic control.

2) **FLAIC for variable force tracking in RUSS:** Building on ℓ_2-FBORL, we design a Fuzzy Learning-based Adaptive Impedance Control (FLAIC) scheme for variable force tracking. FLAIC adaptively adjusts the impedance adaptation rate to reduce overshoot while preserving fast convergence and high tracking accuracy, and, to the best of our knowledge, represents the first ORL-based force tracking framework tailored to clinical RUSS scenarios.

3) **Comprehensive evaluation in simulation, phantom, and in vivo studies:** We perform extensive comparisons between the proposed FLAIC and conventional IC/AIC-based force controllers in high-fidelity simulation, phantom experiments, and in vivo RUSS of upper-limb venous and abdominal scanning. The results demonstrate the superior tracking performance and clinical potential of the proposed approach.

The remainder of this paper is organized as follows. Section II introduces the technical background of IC and RL. Section III elaborates the proposed ℓ_2-FBORL, and its application on variable force tracking, i.e., the FLAIC, and their theoretical analysis in universal approximation and stability performance. In Section IV, simulation and real-world RUSS validation are conducted to compare the performance of FLAIC and conventional methods. At last, a conclusion is given in Section V.

II. Technical Background

A. Impedance Control and Adaptive Impedance Control

IC regulates a desired force-motion relationship as follows:

$$M\ddot{x} + D\dot{x} + K(x - x_{\text{ref}}) = F_{\text{ext}}, \tag{1}$$

where M, D, and K denote the inertia, damping, and stiffness of the virtual impedance; x is the end-effector position; x_{ref} is the reference trajectory; and F_{ext} represents the external force.

AIC extends IC by adjusting parameters to compensate environment uncertainties:

$$M\ddot{x} + \hat{D}\dot{x} + \hat{K}(x - x_{\text{ref}}) = F_{\text{ext}}, \tag{2}$$

where $\hat{D}$ and $\hat{K}$ are the estimated damping and stiffness parameters, which are updated through various adaptation schemes to achieve stable control [10], [12], [13], [15], [16].

B. Markov Decision Processes and Reinforcement Learning

We adopt the standard reinforcement learning (RL) formulation, in which the control problem is cast as a Markov decision process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, d_0, R, \gamma)$. Here, $\mathcal{S}$ and $\mathcal{A}$ denote the state and action spaces, respectively; $T(\cdot \mid \mathbf{s}, a)$ specifies the transition kernel that governs the evolution of the system

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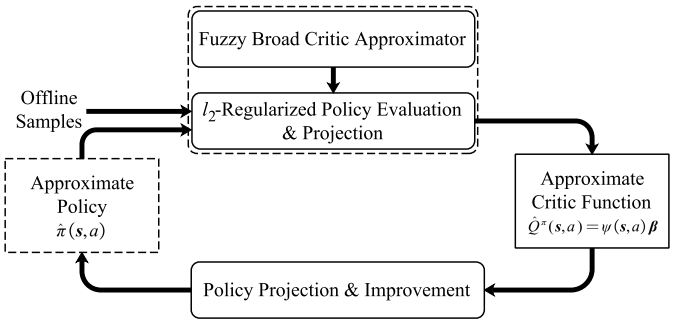


Fig. 2. Flowchart of the proposed ℓ_2-FBORL algorithm.

from state $\mathbf{s} \in \mathcal{S}$ under action $a \in \mathcal{A}$; d_0 is the distribution of the initial state; $R(\mathbf{s}, a) \in \mathbb{R}$ is the immediate reward obtained at $(\mathbf{s}, a)$; and $\gamma \in (0, 1]$ is the discount factor that trades off immediate and future rewards.

A (stochastic) policy $\pi(a|\mathbf{s})$ maps each state to a distribution over actions. Given a policy π, its performance is measured by the expected discounted return

$$J(\pi) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right], \tag{3}$$

and the goal of RL is to identify a policy $\pi^{\star}$ that maximizes $J(\pi)$.

Associated with any policy π is the action-value function (critic)

$$Q^{\pi}(\mathbf{s}, a) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t \;\middle|\; \mathbf{s}_0 = \mathbf{s},\ a_0 = a\right], \tag{4}$$

which satisfies the Bellman expectation equation

$$Q^{\pi}(\mathbf{s}, a) = \mathbb{E}_{\mathbf{s}' \sim T(\cdot|\mathbf{s},a)}\Big[R(\mathbf{s}, a) + \gamma\, \mathbb{E}_{a' \sim \pi(\cdot|\mathbf{s}')} Q^{\pi}(\mathbf{s}', a')\Big]. \tag{5}$$

The optimal critic is $Q^{\star}(\mathbf{s}, a) = \sup_{\pi} Q^{\pi}(\mathbf{s}, a)$, inducing the optimal policy

$$\pi^{\star}(\mathbf{s}) = \arg\max_{a \in \mathcal{A}} Q^{\star}(\mathbf{s}, a). \tag{6}$$

III. Proposed ℓ_2-FBORL and FLAIC for Safe Variable Force Tracking

In this part, the proposed ℓ_2-FBORL is first elaborated, which employs a fuzzy broad critic approximator (FBCA) in an ensemble learning manner of fuzzy systems and broad enhancement learning systems. Based on ℓ_2-FBORL and AIC, a novel fuzzy learning-based variable force tracking approach, i.e., FLAIC, is then proposed. Finally, the universal approximation capability and convergence properties of both ℓ_2-FBORL and FLAIC are analyzed theoretically.

A. Learning Procedure of ℓ_2-FBORL From Imperfect Offline Data

The learning procedure of ℓ_2-FBORL is shown in Fig. 2, and the detailed learning process is described in the following.

(1) Sample Acquisition: The samples used for deriving a policy can be obtained via either online interaction or offline acquisition. In many application domains, one common approach is to execute a random policy to gather the necessary data. Nevertheless, in safety-critical fields such as RUSS in this research, performing online sampling with random policies may lead to medical safety concerns. To address this issue, offline data from stabilizing controllers or expert demonstrations are collected for policy learning. Therefore, a dataset $\mathcal{D} = \left\{\left(\boldsymbol{s}_t, a_t, r_t, \boldsymbol{s}_t'\right) \mid t = 1, 2, \ldots, N\right\}$ in the MDP format is constructed for the training of the ORL policy. Within the MDP framework, when an action a_t in state $\boldsymbol{s}_t$ is executed, it obtains a reward r_t and the system evolves to the subsequent state $\boldsymbol{s}_t'$ according to the environment dynamics.

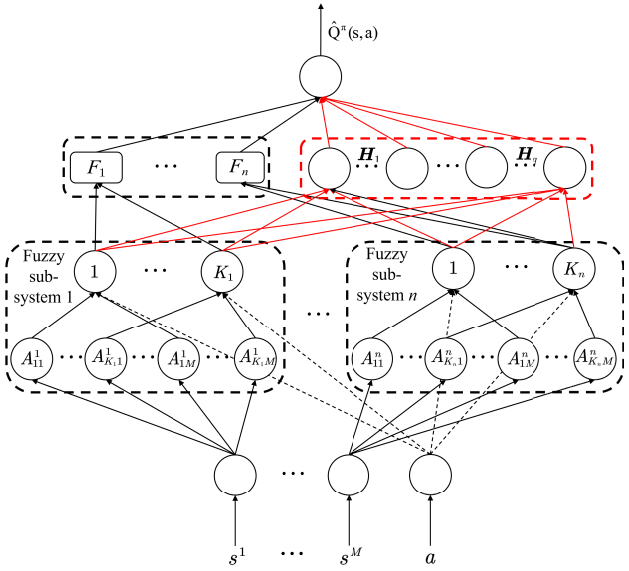


Fig. 3. Fuzzy broad critic approximator. Black lines indicate the ensemble fuzzy learning process, and red lines indicate the broad enhancement learning process.

(2) Fuzzy Sub-System Ensemble Learning: The FBCA can be regarded as an ensemble learning architecture composed of n fuzzy sub-systems together with m enhancement learning modules, corresponding to m groups of enhancement nodes in the FBLS framework, as shown in Fig. 3. Let a state–action pair $(\boldsymbol{s}, a)$ with $\boldsymbol{s} = (s^1, s^2, \cdots, s^M)$ denote one input sample, and let $Q(\boldsymbol{s}, a)$ represent the associated state–action critic value estimated by the FBCA. The i-th fuzzy sub-system contains K_i fuzzy rules.

In the fuzzy sub-system design of FBCA, directly concatenating the state $\boldsymbol{s}$ and action a for fuzzy feature extraction does not sufficiently emphasize the role of the action. Inspired by [18] and [32], we adopt a two-stage fuzzy processing strategy: fuzzy features are first extracted from $\boldsymbol{s}$, after which the action a is incorporated into each fuzzy sub-system via Kronecker encoding. The fuzzifier transforms a crisp input state $\boldsymbol{s} \in \mathcal{S} \subset \mathbb{R}^M$ into fuzzy representations over $\mathcal{S}$, where each fuzzy set is characterized by a membership function $\mu_F : \mathcal{S} \to [0, 1]$. In this work, a singleton fuzzifier is employed.

A fuzzy rule base consists of linguistic IF–THEN rules. Specifically, the rule base $\mathcal{R}_i$ of the i-th fuzzy sub-system contains K_i rules of the form

$$\mathcal{R}_i:\ \textbf{IF}\ s^1 \text{ is } \mathcal{A}_{k1}^i\ \textbf{and}\ \cdots\ \textbf{and}\ s^M \text{ is } \mathcal{A}_{kM}^i,\quad \textbf{THEN}\ z_k^i = \mathcal{B}_k^i,$$

where s^p $(p = 1, 2, \ldots, M)$ denote the input variables, and z_k^i is the rule output. Each rule defines a fuzzy implication

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$\mathcal{A}^i_{k1} \times \cdots \times \mathcal{A}^i_{kM} \to \mathcal{B}^i_k$, with the membership degree computed using standard product or minimum operators [33].

The consequent z^i_k is modeled by a fuzzy basis function (FBF). In this study, a first-order Takagi–Sugeno (TS) fuzzy model [34] is adopted,

$$z^i_k = f^i_k(\boldsymbol{s}) = \sum_{l=1}^{M} \alpha^i_{kl} s^l, \tag{7}$$

where the coefficients α^i_{kl} are randomly generated from a uniform distribution on $[0, 1]$, following a procedure similar to that used in BLS.

The fuzzy inference engine establishes a mapping between fuzzy sets in the input domain $\mathcal{S}$ and those in the output domain $\mathcal{R}$. Under the rule base $\mathcal{R}_i$, the inferred fuzzy set is obtained through the sup-star composition

$$\mu_{\mathcal{A}_s \circ \mathcal{R}_i}(z^i_k) = \sup_{\mathring{s} \in \mathcal{S}} \left[\mu_{\mathcal{A}_s}(\mathring{s}) * \prod_{p=1}^{M} \mu_{\mathcal{A}^i_{kp}}(\mathring{s}^p) * \mu_{\mathcal{B}^i_k}(z^i_k) \right]. \tag{8}$$

In this work, Gaussian membership functions are adopted for the antecedent fuzzy sets,

$$\mu^i_{kp}(s^p) = \exp\left(- \left(\frac{s^p - c^i_{kp}}{\sigma^i_{kp}} \right)^2 \right), \tag{9}$$

where c^i_{kp} and σ^i_{kp} denote the center and width, respectively.

The normalized activation level (weighted fire strength) of the k-th rule is given by

$$\lambda^i_k = \frac{\mu_{\mathcal{A}_s \circ \mathcal{R}_i}(z^i_k)}{\sum_{k=1}^{K_i} \mu_{\mathcal{A}_s \circ \mathcal{R}_i}(z^i_k)}. \tag{10}$$

The weighted outputs of all fuzzy rules are aggregated into an intermediate feature vector

$$\boldsymbol{Z}_i = \left[\lambda^i_1 z^i_1, \lambda^i_2 z^i_2, \ldots, \lambda^i_{K_i} z^i_{K_i} \right], \tag{11}$$

which corresponds to the i-th group of feature nodes in the BLS paradigm. Concatenating all n fuzzy sub-systems yields

$$\boldsymbol{Z} = (\boldsymbol{Z}_1, \boldsymbol{Z}_2, \ldots, \boldsymbol{Z}_n) \in \mathbb{R}^{\sum_{i=1}^{n} K_i}. \tag{12}$$

Denoting the mapping from $\boldsymbol{s}$ to $\boldsymbol{Z}_i$ as $\phi_i(\boldsymbol{s})$, the state–action feature vector is constructed via one-hot action encoding,

$$\phi_i(\boldsymbol{s}, a) = \boldsymbol{e_I}(a) \otimes \phi_i(\boldsymbol{s}), \tag{13}$$

where $\otimes$ denotes the Kronecker product. This design avoids direct state–action concatenation and is particularly suitable for RL settings.

Finally, the defuzzified output of the i-th fuzzy sub-system is computed as

$$F_i = \phi_i(\boldsymbol{s}, a)^T \boldsymbol{\omega}_i, \tag{14}$$

and the outputs of all fuzzy sub-systems are aggregated as

$$F^{(n)} = \sum_{i=1}^{n} F_i = \sum_{i=1}^{n} \phi_i(\boldsymbol{s}, a)^T \boldsymbol{\omega}_i. \tag{15}$$

(3) Broad Enhancement Learning Systems: The intermediate matrix $\boldsymbol{Z}$ is forwarded to the enhancement learning systems for nonlinear transformation. Assume each enhancement learning system contains m neurons; then the output of the j-th enhancement group is given by

$$\boldsymbol{H}_j = \xi_j \left(\boldsymbol{Z} \boldsymbol{W}_{hj} + \boldsymbol{\beta}_{hj} \right) \in \mathbb{R}^{N \times m}, \tag{16}$$

where $\boldsymbol{W}_{hj}$ and $\boldsymbol{\beta}_{hj}$ are randomly initialized weights and biases in $[0, 1]$ connecting the fuzzy sub-systems to the corresponding enhancement nodes. The activation function $\xi_j(\cdot)$ is a nonconstant, piecewise continuous function, ensuring the FBLS universal approximation property. This nonlinear transformation acts as a complementary component to the first-order fuzzy systems. Let $\varphi_j(\boldsymbol{s}) = \xi_j\big(\Phi(\boldsymbol{s})\big)$, where $\varphi_j(\cdot)$ serves as a basis function representing the constructed features of $\boldsymbol{s}$ in the enhancement layer. The state features of the enhancement learning systems can then be extended to state–action features as

$$\varphi_j(\boldsymbol{s}, a) = \boldsymbol{e_I}(a) \otimes \varphi_j(\boldsymbol{s}), \tag{17}$$

where $\otimes$ denotes the Kronecker product, and $\boldsymbol{e_I}(a)$ is the one-hot encoding vector corresponding to action a.

Finally, the ensemble outputs of the fuzzy sub-systems ($F^{(n)}$), together with the enhancement layer outputs ($\boldsymbol{H} = [\boldsymbol{H}_1, \cdots, \boldsymbol{H}_m]$), are used to compute the overall output of the $\hat{Q}(\boldsymbol{s}, a)$. Let $\boldsymbol{W}_h = [\boldsymbol{w}^1, \boldsymbol{w}^2, \ldots, \boldsymbol{w}^m] \in \mathbb{R}^{m \times 1}$ denote the weight matrix connecting the enhancement layer to the top layer. Then, the FBLS output for a state–action pair $(\boldsymbol{s}, a)$ is given by

$$\begin{aligned} \hat{Q}(\boldsymbol{s}, a) &= F^{(n)} + \sum_{j=1}^{m} \varphi_j(\boldsymbol{s}, a)^T \boldsymbol{w}^j \\ &= \sum_{i=1}^{n} \phi_i(\boldsymbol{s}, a)^T \boldsymbol{\omega}_i + \sum_{j=1}^{m} \varphi_j(\boldsymbol{s}, a)^T \boldsymbol{w}^j \\ &= \psi(\boldsymbol{s}, a) \boldsymbol{\omega}^m_n \end{aligned} \tag{18}$$

where $\psi_k(\boldsymbol{s}, a) = \begin{cases} \phi_k(\boldsymbol{s}, a), & if\ k \le n \\ \varphi_{k-n}(\boldsymbol{s}, a), & if\ k > n \end{cases}$, $\psi(\boldsymbol{s}, a) = \left[\psi_1(\boldsymbol{s}, a), \psi_2(\boldsymbol{s}, a), \cdots, \psi_{m+n}(\boldsymbol{s}, a) \right]^T$, and $\boldsymbol{\omega}^m_n = \left[\boldsymbol{\omega}_1, \cdots, \boldsymbol{\omega}_n, \boldsymbol{\omega}^1 \cdots, \boldsymbol{\omega}^m \right]$.

Remark 1: From the perspective of BLS theory, FBCA can be viewed as a system comprising n groups of feature nodes processed by n fuzzy sub-systems and m groups of enhancement nodes. From the perspective of fuzzy learning systems, FBCA can be viewed as an ensemble learning system that integrates n fuzzy sub-systems and m enhancement learning systems. To further enhance its nonlinear feature representation capabilities of fuzzy sub-systems, FBCA draws inspiration from BLS by introducing enhancement learning systems, which significantly improve the nonlinear approximation capabilities of the fuzzy learning system. In summary, FBCA is a linear state-action critic approximator and it is the first trial of FBLS in ORL, particularly in the domain of robotic ORL control.

(4) Policy Learning with ℓ_2-**Regularization:** Based on the pre-collected samples in the format of $\mathcal{D} = \left\{ \left(\boldsymbol{s}_t, a_t, r_t, \boldsymbol{s}'_t \right) \mid t = 1, 2, \cdots, N \right\}$, we define the following sample matrices:

$$\begin{aligned} \tilde{\Phi} &\equiv \left[\psi(s_1, a_1), \cdots, \psi(s_N, a_N) \right]^T, \\ \tilde{\Phi}' &\equiv \left[\psi\left(s'_1, \pi\left(s'_1 \right) \right), \cdots, \psi\left(s'_N, \pi\left(s'_N \right) \right) \right]^T, \\ \tilde{R} &\equiv [r, \cdots, r_N]^T. \end{aligned} \tag{19}$$

Leveraging the sample efficiency of the least-squares temporal-difference (LSTD) learning algorithm [35], we employ it to

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obtain a data-driven approximation of the FBCA, and thus the corresponding weight vector of $\hat{Q}(s,a)$ is then computed as

$$\boldsymbol{\omega}_n^m = \tilde{f}(\boldsymbol{\omega}_n^m) = \arg\min_{u\in R^K} \left\| \widetilde{\Phi} u - \left(\widetilde{R} + \gamma \widetilde{\Phi}' \boldsymbol{\omega}_n^m\right) \right\|^2. \tag{20}$$

In clinical practice, large-scale MDP samples are typically hard to collect. The data scarcity exposes traditional LSTD to over-fitting. Consequently, we adopt an ℓ_2–regularized variant that remains effective in the low-sample regime.

Let $\beta \geq 0$ denote the regularization coefficient, and the objective function is transformed into

$$\tilde{f}(\boldsymbol{\omega}_n^m) = \arg\min_{u\in\mathbb{R}^K} \tfrac{1}{2} \left\| \tilde{\Phi} u - (\tilde{R} + \gamma \tilde{\Phi}' \boldsymbol{\omega}_n^m) \right\|^2 + \tfrac{\beta}{2} \|u\|_2^2, \tag{21}$$

which remains convex and differentiable, and the closed-form minimizer is

$$u^\star = (\tilde{\Phi}^\top \tilde{\Phi} + \beta I)^{-1} \tilde{\Phi}^\top (\tilde{R} + \gamma \tilde{\Phi}' \boldsymbol{\omega}_n^m). \tag{22}$$

Enforcing the fixed-point condition $\boldsymbol{\omega}_n^m = u^\star$ yields the regularized linear system

$$\boldsymbol{\omega}_n^m = \left(\tilde{\Phi}^\top (\tilde{\Phi} - \gamma \tilde{\Phi}') + \beta I\right)^{-1} \tilde{\Phi}^\top \tilde{R} = (\tilde{A} + \beta I)^{-1} \tilde{b}, \tag{23}$$

where $\tilde{A} \triangleq \tilde{\Phi}^\top (\tilde{\Phi} - \gamma \tilde{\Phi}')$ and $\tilde{b} \triangleq \tilde{\Phi}^\top \tilde{R}$. Since $\tilde{A} + \beta I > 0$ for any $\beta > 0$, the solution exists with probability one even when sample size is small.

(5) Policy Improvement: The policy is improved by applying a greedy operation on the approximate critic function $\hat{\boldsymbol{Q}}^\pi$ over all possible actions $a \in \mathcal{A}$:

$$\pi(\boldsymbol{s}) = \arg\max_{a\in\mathcal{A}} \psi(\boldsymbol{s}, a)\, \boldsymbol{\omega}_n^m. \tag{24}$$

The policy $\pi(\boldsymbol{s})$ is updated iteratively to improve its control performance. After each iteration, the matrix $\tilde{\Phi}'$ in (19) is updated based on the newly learned policy. After each iteration, the updated weight vector $(\boldsymbol{\omega}_n^m)'$ is compared with the previous weight vector $\boldsymbol{\omega}_n^m$. If the change satisfies the convergence criterion $\left\| \boldsymbol{\omega}_n^m - (\boldsymbol{\omega}_n^m)' \right\|_2 < \varepsilon$, where ε is a predefined small threshold, the iteration process is terminated.

Overall, the procedure of the proposed ℓ_2-FBORL is shown in Algorithm 1. After that, we can develop a learning-based robotic variable force tracking methods based on the proposed ℓ_2-FBORL method to achieve high control precision in dynamic unstructured environments, especially in biomechanical unstructured environment.

Overall, the complete procedure of the proposed ℓ_2-FBORL algorithm is summarized in Algorithm 1. Upon convergence, the learned policy provides an optimal mapping from the system state to the control action under the offline data distribution, and can be directly embedded as a high-level decision module. Based on the learned policy and critic structure obtained from ℓ_2-FBORL, a learning-based robotic variable force tracking framework can be further developed. In particular, the learned policy is integrated with a safe adaptive impedance control scheme, enabling precise and stable force regulation in dynamic and unstructured environments, especially in robotic US scanning scenarios.

Algorithm 1 ℓ_2-FBORL

Input: γ: The discount factor; ε: Termination factor;
n: Number of fuzzy sub-systems;
m: Number of broad enhancement learning systems;
T_M: Maximum iteration number
Output: $\boldsymbol{\omega}_n^m$: The output weights

Initialization: $\tilde{\boldsymbol{A}} \leftarrow 0$, $\tilde{\boldsymbol{b}} \leftarrow 0$, $\boldsymbol{\omega}_n^m \leftarrow 0$, $\left(\boldsymbol{\omega}_n^m\right)' \leftarrow 0$, $i \leftarrow 0$, $\delta \leftarrow 1/10^5$;
Collecting Samples: Construct a dataset D with N samples, $D = \left\{\left(\boldsymbol{s}_t, a_t, r_t, \boldsymbol{s}_t'\right) \mid t = 1, 2, \cdots, N\right\}$;
for each sample $\left(\boldsymbol{s}_t, a_t, r_t, \boldsymbol{s}_t'\right) \in D$ **do**
  for $i \in [1, n]$ **do**
    $\phi_i(\boldsymbol{s}) = \left[\lambda_1^i z_1^i, \lambda_2^i z_2^i, \ldots, \lambda_{K_i}^i z_{K_i}^i\right]$
    $\phi_i(\boldsymbol{s}, a) = \boldsymbol{e_I}(a) \otimes \phi_i(\boldsymbol{s})$
  end for
  for $i \in [1, m]$ **do**
    $\varphi_j(\boldsymbol{s}) = \xi_j\left(\left[(\phi_1(\boldsymbol{s}), \phi_2(\boldsymbol{s}), \ldots, \phi_n(\boldsymbol{s}))\right] \boldsymbol{W}_{hj} + \boldsymbol{\beta}_{hj}\right)$
    $\varphi_j(\boldsymbol{s}, a) = \boldsymbol{e_I}(a) \otimes \varphi_j(\boldsymbol{s})$
  end for
end for
$\pi^0(\boldsymbol{s}) \leftarrow random()$, $\boldsymbol{\omega}_n^m \leftarrow \left(\boldsymbol{\omega}_n^m\right)'$;
Update $\boldsymbol{\Psi}$, $\boldsymbol{\Psi}'$ and $\boldsymbol{R}$ according to Eq. (19);
repeat
  $\tilde{\boldsymbol{A}} = \boldsymbol{\Psi}^T\left(\boldsymbol{\Psi} - \gamma \boldsymbol{\Psi}'\right)$, $\tilde{\boldsymbol{b}} = \boldsymbol{\Psi}^T \boldsymbol{R}$;
  $\left(\boldsymbol{\omega}_n^m\right)' \leftarrow (\tilde{\boldsymbol{A}} + \delta \mathbf{I})^{-1} \tilde{\boldsymbol{b}}$;
  $\pi^{v+1}(\boldsymbol{s}) = \arg\max_{\boldsymbol{a}\in\mathcal{A}} \psi(\boldsymbol{s}, \boldsymbol{a}) \left(\boldsymbol{\omega}_n^m\right)'$;
  $v = v + 1$;
until $\left\| \boldsymbol{\omega}_n^m - \left(\boldsymbol{\omega}_n^m\right)' \right\|_2 < \varepsilon$ OR $v > T_M$
return $\boldsymbol{\omega}_n^m$.

B. Safe Learning-Based Adaptive Impedance Control for Robotic Variable Force Tracking

In the context of robotic variable force tracking, we focus exclusively on the control force along the Z-axis of the force sensor. The desired impedance behavior is formulated as

$$m_d \ddot{\Delta x}_r + b_d \dot{\Delta x}_r + k_d \Delta x_r - f_m + f_d = 0, \tag{25}$$

where m_d, b_d, and k_d are the desired inertia, damping, and stiffness, respectively; f_m and f_d are the measured and desired contact forces; $\Delta x_r = x_e - x$ is the position error between the estimated probe position x_e and the target position x.

The environment is modeled as a position-dependent stiffness system to capture tissue dynamics [16]:

$$f_{\text{env}} = k_{\text{env}}(x_m - x_{\text{env}}), \tag{26}$$

where k_{env} and x_{env} represent the environment stiffness and its rest position, respectively.

Applying a z-transform to the trajectory error Δx_r yields the frequency-domain representation [12]:

$$m_d z^2 \Delta x_r(z) + b_d z \Delta x_r(z) + k_d \Delta x_r(z) = \Delta f(z), \tag{27}$$

where z is the complex frequency variable.

The resulting force-position dynamics can be expressed through the transfer function $G(z) = [m_d z^2 + b_d z + k_d]^{-1}$, which maps the force tracking error Δf to the position correction Δx_r. Consequently, the reference trajectory is adjusted as $x_d = x_r + \Delta x_r = x_r + G(z)\Delta f$. Under position servocontrol, $x_m = x_d$, and the measured force satisfies $f_m = f_{\text{env}}$. Therefore, the force tracking error becomes

$$\begin{aligned} \Delta f(z) &= f_m - f_d = k_{\text{env}}(x_d - x_{\text{env}}) - f_d \\ &= k_{\text{env}}\left(x_r + G(z)\Delta f(z) - x_{\text{env}}\right) - f_d. \end{aligned} \tag{28}$$

Substituting $G(z) = [m_d z^2 + b_d z + k_d]^{-1}$ into (28) gives

$$\Delta f(z) = \frac{(m_d z^2 + b_d z + k_d)\left[k_{\text{env}}(x_r - x_{\text{env}}) - f_d\right]}{m_d z^2 + b_d z + k_d - k_{\text{env}}}. \tag{29}$$

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The corresponding steady-state error is

$$\Delta f_{ss} = \lim_{z\to 0} \Delta f(z) = \frac{k_d\left[k_{\text{env}}(x_r - x_{\text{env}}) - f_d\right]}{k_d - k_{\text{env}}}. \tag{30}$$

Zero steady-state error requires either $k_{\text{env}}(x_r - x_{\text{env}}) - f_d = 0$ or $k_d = 0$. Since k_{env} and x_{env} are generally unknown, setting $k_d = 0$ provides an equilibrium with $\Delta f_{ss} = 0$.

The adaptive impedance control (AIC) along the Z-axis operates in two phases: free-space control and contact-space control. Free-space motion corresponds to the probe moving without contacting tissue, whereas contact-space control handles both initial skin contact and variable force tracking (see Fig. 1(a)). The AIC dynamics can be described as,

$$\begin{cases} m_d\ddot{\Delta x}_r + b_d\dot{\Delta x}_r + k_d\Delta x_r = -f_d, \text{(free space)} \\ m_d\ddot{\Delta x}_r + b_d(\dot{\Delta x}_r + \Xi) = f_e - f_d, \text{(contact space)} \end{cases} \tag{31}$$

where Ξ is a time-varying compensation term updated as

$$\Xi(t) = \Xi(t-\lambda) + \eta\frac{f_d(t-\lambda) - f_e(t-\lambda)}{b_d}, \quad \eta > 0, \tag{32}$$

with η as the adaptation rate and λ the sampling period.

The damping ratio of the compensated AIC controller is

$$\zeta = \frac{b_d}{2\sqrt{m_d(1+\eta)k_e}}, \tag{33}$$

where k_e denotes the unknown tissue stiffness. Notably, larger η reduces ζ, increasing the risk of oscillation and resulting in patient discomfort and even medical accident. To maintain safe and precise force tracking, we propose a learning-based strategy, FLAIC, leveraging the ℓ_2-FBORL framework to adaptively regulate η and minimize force overshoot to maintain clinical safety when RUSS applied.

In the proposed FLAIC framework, the MDP state is defined as $\boldsymbol{s} = [\delta f, \dot{\delta f}]$, where $\delta f = f_e - f_d$ denotes the force tracking error and $\dot{\delta f}$ represents its temporal variation. The action space is discretized into a finite set of candidate update rates η, $\mathcal{A} = \{a^0, a^1, a^2, a^4\}$. The reward function is designed to promote accurate and stable force tracking while suppressing overshoot. In robotic ultrasound scanning, force deviations within ± 0.5 N are considered acceptable, i.e., $|\delta f| \le 0.5$, and yield a positive reward ($r = 1$). When the force error exceeds this bound but evolves toward the target, i.e., $|\delta f| > 0.5$ and $\text{sgn}(\delta f \cdot \delta f_d) \le 0$, a moderate penalty is applied, $r = -\alpha_1\delta f - \alpha_2\dot{\delta f}$, to discourage slow convergence. In contrast, force overshoot, characterized by $|\delta f| > 0.5$ and $\text{sgn}(\delta f \cdot \delta f_d) > 0$, incurs an additional large penalty, $r = -\alpha_1\delta f - \alpha_2\dot{\delta f} - 100$.

Finally, the procedure for variable force tracking in robotic US scanning is implemented collaboratively by the robot operator and the RUSS system. The workflow is as follows:

1) Communication with the patient to localize the symptomatic region, performed by operators.
2) Determination of the patient-specific maximum allowable scanning forces, denoted as F_{Non} and F_{Sym}, and also the symptomatic area through palpation, performed by operators.
3) Assignment of distinct target forces to different segments of the scanning trajectory during path planning, executed by the RUSS system.
4) Execution of variable force tracking along the planned trajectory using the learned FLAIC controller via the RUSS system.

The overall training and deployment workflow of the FLAIC controller within the RUSS framework is illustrated in Fig. 4.

C. Performance Analysis

Let $Q^*(\boldsymbol{s}, a) \in C(\mathbf{I}^d)$ denote the optimal state–action critic, where $\boldsymbol{s} \in \mathbf{I}^d = [0,1]^d$ and $a \in \mathcal{A}$ is discrete. The FBLS-like critic approximator can be decomposed as

$$\hat{Q}^{\hat{\pi}}(\boldsymbol{s}, a) = f_{be}(\boldsymbol{s}, a) + f_{bh}(\boldsymbol{s}, a), \tag{34}$$

where f_{be} is the FBF ensemble subsystem and f_{bh} is the broad enhancement (RVFL) subsystem.

For any compact $\mathbf{K} \subset \mathbf{I}^d$, define the approximation error by

$$\rho_{\mathbf{K}}(Q^*, \hat{Q}^{\hat{\pi}}) = \left(\mathbb{E}\left[\int_{\mathbf{K}} \sum_{a\in\mathcal{A}} \left(Q^*(\boldsymbol{s}, a) - \hat{Q}^{\hat{\pi}}(\boldsymbol{s}, a)\right)^2 d\boldsymbol{s}\right]\right)^{1/2}, \tag{35}$$

where $\mathbb{E}[\cdot]$ is taken over the randomness in the approximator (if any).

Lemma 1: [33]: For any continuous function g on $\mathbf{K}$ and any $\varepsilon > 0$, there exists an FBF expansion $f_{be} \in Y$ such that $\sup_{\boldsymbol{x}\in\mathbf{K}} |g(\boldsymbol{x}) - f_{be}(\boldsymbol{x})| < \varepsilon$. Thus, $\rho_{\mathbf{K}}(g, f_{be}) < \varepsilon$.

Lemma 2: [36]: Given any continuous $f \in C(\mathbf{I}^d)$ on $\mathbf{K}$, let f_{bh} be an RVFL approximator and $f_{rn} = f - f_{bh}$. Then for any $\varepsilon > 0$, there exists f_{cn} such that $\rho_{\mathbf{K}}(f_{cn}, f_{rn}) < \varepsilon$.

Applying Lemmas 1–2 to $\hat{Q}^{\hat{\pi}} = f_{be} + f_{bh}$, we obtain

$$\rho_{\mathbf{K}}(Q^*, \hat{Q}^{\hat{\pi}}) \le \rho_{\mathbf{K}}(g, f_{be}) + \rho_{\mathbf{K}}(f_{cn}, f_{rn}) < 2\varepsilon. \tag{36}$$

Hence,

$$\lim_{m,n\to\infty} \rho_{\mathbf{K}}(Q^*, \hat{Q}^{\hat{\pi}}) = 0, \tag{37}$$

which establishes the universal approximation capability of the FBLS-based critic.

Remark 2: Consider a FBLS-based critic network with finite FBF feature layers and broad enhancement layers. Owing to the universal approximation property of fuzzy sub-systems coupled with broad RVFL architectures, such a structure—whose internal parameters are merely initialized randomly within a bounded interval—can approximate the optimal critic Q^* to any prescribed accuracy $\varepsilon > 0$.

Remark 3: Unlike supervised learning, RL lacks ground-truth labels for each state–action pair. The Bellman equation (5) circumvents this deficit by recasting the critic-estimation task as an ordinary supervised regression problem. Consequently, the output weights of ℓ_2-FBORL (and FLAIC) are obtained through a closed-form ℓ_2-regularized least-squares solution.

Leveraging the above approximation guarantee, ℓ_2-FBORL converges to a fixed weight vector whose deviation from the optimal critic is controlled by the network width. Following [18], the least-squares fixed-point iteration with a linear FBLS-based critic yields

$$\forall k \ge 1, \quad \|\hat{Q}^{\pi_k} - Q^{\pi_k}\|_\infty \le \epsilon, \tag{38}$$

and thus

$$\limsup_{k\to\infty} \|\hat{Q}^{\pi_k} - Q^*\|_\infty \le \frac{2\gamma\epsilon}{(1-\gamma)^2}. \tag{39}$$

Thus, ℓ_2-FBORL is a provably stable and theoretically well-founded algorithm.

Architecturally, ℓ_2-FBORL strictly preserves the FBLS architecture, which contains fuzzy feature nodes and enhancement nodes. Moreover, both of them employ the least squares

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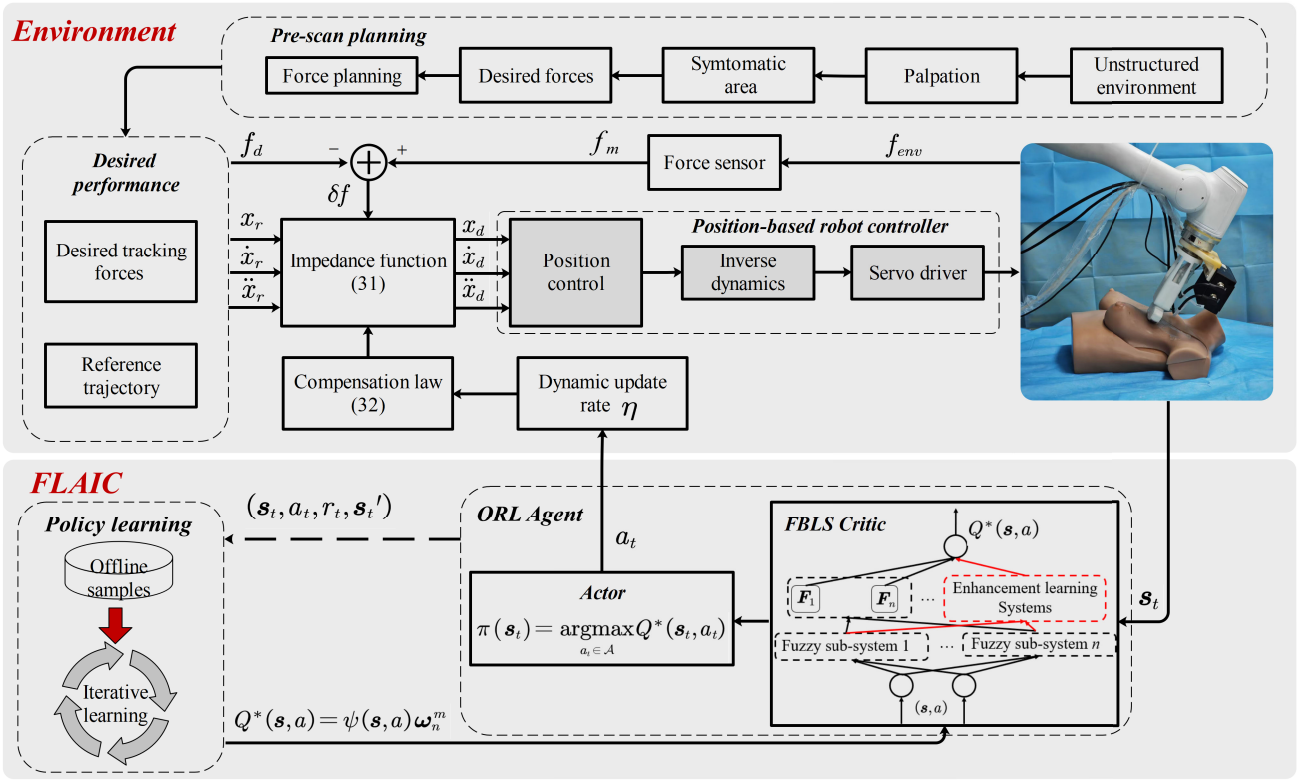


Fig. 4. FLAIC control framework.

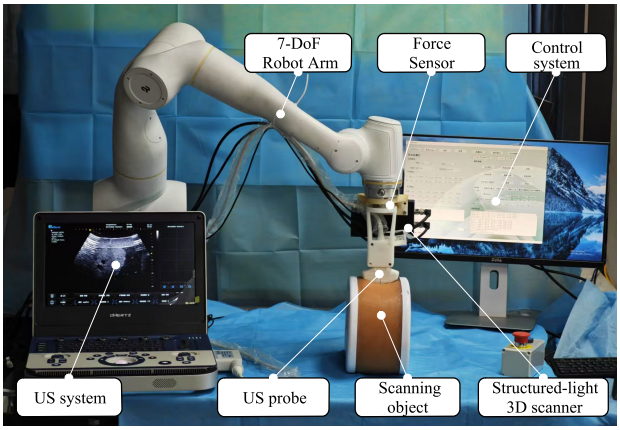


Fig. 5. Developed robot system for real-world validation.

technique to learn their weight vector. Consequently, ℓ_2-FBORL represents the first principled transposition of FBLS into the ORL regime, thereby conferring FBLS's advantages for robotic force control in clinical scenarios, and the FLAIC is also the first FBLS-based ORL method for robotic variable force tracking scenarios.

IV. Experiments and Results Analysis

A. RUSS System Construction

RUSS provide an effective means to enhance the consistency of ultrasound examinations while alleviating the operational burden on clinicians. In a typical RUSS workflow, the scanning task is planned in advance according to patient-specific anatomical and mechanical properties. Such planning can be performed using depth-enhanced visual information obtained from RGB-D sensors [37] or preoperative medical images such as CT or MRI [38]. During execution, the ultrasound probe is commanded to exert prescribed normal forces onto the target tissue, as illustrated in Fig. 1(a). To validate the proposed control strategy, a complete RUSS experimental platform was developed, as shown in Fig. 5. The system employs a 7-degree-of-freedom medical-grade robotic manipulator (Diana7-Med, Agile Robots, Germany) as the motion execution unit, which is programmed in C++ under the Qt framework. Ultrasound image streams are acquired from an imaging device (eHertzR Express, sSonic, China) through a dedicated image capture card (VGA41A-41 +, JoinHope Image Inc., China), delivering frames at a resolution of 1000 × 800 pixels and a sampling rate of 20 Hz. A convex-array ultrasound probe (C5-1Xp, sSonic, China) is rigidly attached to the robot end-effector via a customized fixture, which also incorporates a six-axis force/torque sensor (Nano Mini40, ATI Industrial Automation, USA) to enable real-time sensing of interaction forces. All computation and system coordination are performed on a workstation equipped with an NVIDIA RTX 3080 GPU, an Intel i7-12700 CPU operating at 2.1 GHz, and 64 GB of RAM.

B. Demonstration Sample Collection and Robotic Policy Learning

The sample set used to extract policies can be collected via an online or offline approach. In general, the sample set can be collected via a random policy in a specific domain. However, in domains like robots, sampling online with random policies will lead to safety problem, especially in our clinical scenarios.

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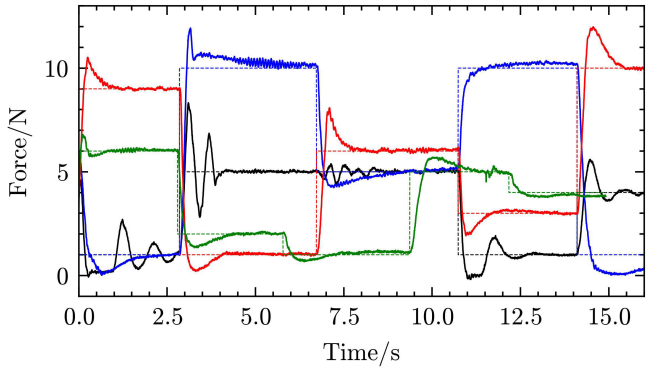


Fig. 6. Demonstration samples collected via the developed RUSS system. Different colors represent different RUSS trajectories.

Algorithm 2 RUSS MDP Data Collection

Initialize sample set $D = \{\}$;
Define symptomatic region via tactile palpation;
for each episode $k = 1, \ldots, N_{ep}$ **do**
  Randomly generate target force $f_{\text{target}} \in [0, 10]$;
  Randomly choose mode $\leftarrow$ {random, fixed};
  if mode = fixed **then**
    Select an action $a_{\text{fixed}} \in \mathcal{A}$ for a whole scan;
  end if
  Initialize $t = 0$, state $s_0 = [\delta f_0, \delta f_0']$;
  repeat
    if mode = random **then**
      Randomly select action $a_t \in \mathcal{A}$;
    else
      $a_t \leftarrow a_{\text{fixed}}$;
    end if
    Execute a_t and wait one control step;
    Compute reward r_t according to Eq.-DEL (38);
    $D \leftarrow D \cup (s_t, a_t, r_t, s_t')$;
    Update s_{t+1} from ATI force sensor feedback;
    $t \leftarrow t + 1$;
  until (end of scan trajectory)
end for
return the sample set D.

Consequently, stabilizing controllers via demonstrations or a near-optimal policy are adopted in these cases to collect demonstration data.

In this research, we collected samples in the MDP format on a phantom to ensure safety via stabilizing controllers. Considering the relatively complex structure of the breast phantom (Phantom S230.52, Gaumard Inc., USA), it was adopted for data acquisition. 50 different trajectories with random target forces limited to 10 N based on clinical requirements, are executed via the RUSS system to collect the MDP dataset, comprising approximately 15,000 samples. In practical RUSS operations, the action is typically fixed, i.e., AIC control strategies with fixed η in Eq. (32)). During data collection, we adopted a mixed strategy: with 50% probability, a constant-action scan was performed (η remains constant), and with 50% probability, each control step of the robot executed a randomly selected action from action space $\mathcal{A} = [0.001, 0.005, 0.01, 0.05]$. The detailed strategy is illustrated in Algorithm 2. Representative contact force data from several robotic US scanning trajectories are shown in Fig. 6. Four different colors correspond to four robotic US scanning trials, with the target force indicated by dashed lines of the same color. As observed, these data exhibit either significant overshoot or slow tracking, far from the near-optimal performance. These samples will serve as the basis for learning a near-optimal FLAIC policy.

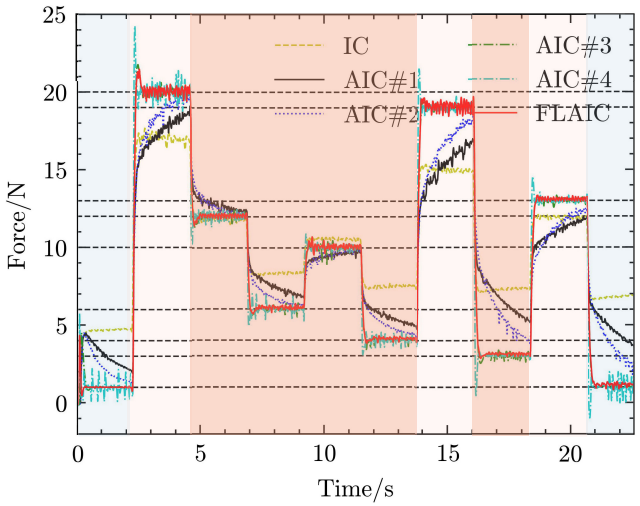


Fig. 7. Robotic variable force tracking in CoppeliaSim simulation.

The proposed FLAIC algorithm is trained utilizing the data acquired above, under the training procedure depicted in Fig. 2. Convergence was achieved after six iterations, satisfying the preset criterion ε of weight variation being less than 10^{-5}. The resultant weights were subsequently applied to a simulation scenario and also our developed RUSS system as shown in Fig. 5.

C. CoppeliaSim Simulation Studies

CoppeliaSim [39] is a high-fidelity, cross-platform simulator, which is widely adopted for robotic research in complicated dynamic application scenarios. The simulation experiment is first conducted via CoppeliaSim to validate the performance of FLAIC controls. To ensure simulation fidelity, the virtual environment is meticulously aligned with the physical RUSS setup: the discrete-time control step is set to 5×10^{-3}s, and the Bullet engine governs rigid-body dynamics and collision detection. For the baseline AIC controller, the Cartesian impedance parameters are $K_d = 1000\,\text{N} \cdot \text{m}^{-1}$, $D_d = 50\,\text{N} \cdot \text{s} \cdot \text{m}^{-1}$, and $M_d = 1\,\text{kg} \cdot \text{m}^2$.

In simulation, we designed a more aggressive force-tracking scenario for stress-test of the proposed method. The reference force is commanded to vary within an even shorter time span, and its peak magnitude is doubled from 10 N (the limit used in real-world applications) to 20 N, so as to stress-test the proposed FLAIC under extreme conditions. To provide a comprehensive benchmark, we compare FLAIC with four AIC variants (i.e., AIC#1 ($\eta = 0.001$), AIC#2 ($\eta = 0.005$), AIC#3 ($\eta = 0.01$), and AIC#4 ($\eta = 0.05$)) and the conventional IC method (i.e., $\eta = 0$). The corresponding results are demonstrated in Fig. 7. The results reveals that IC actually

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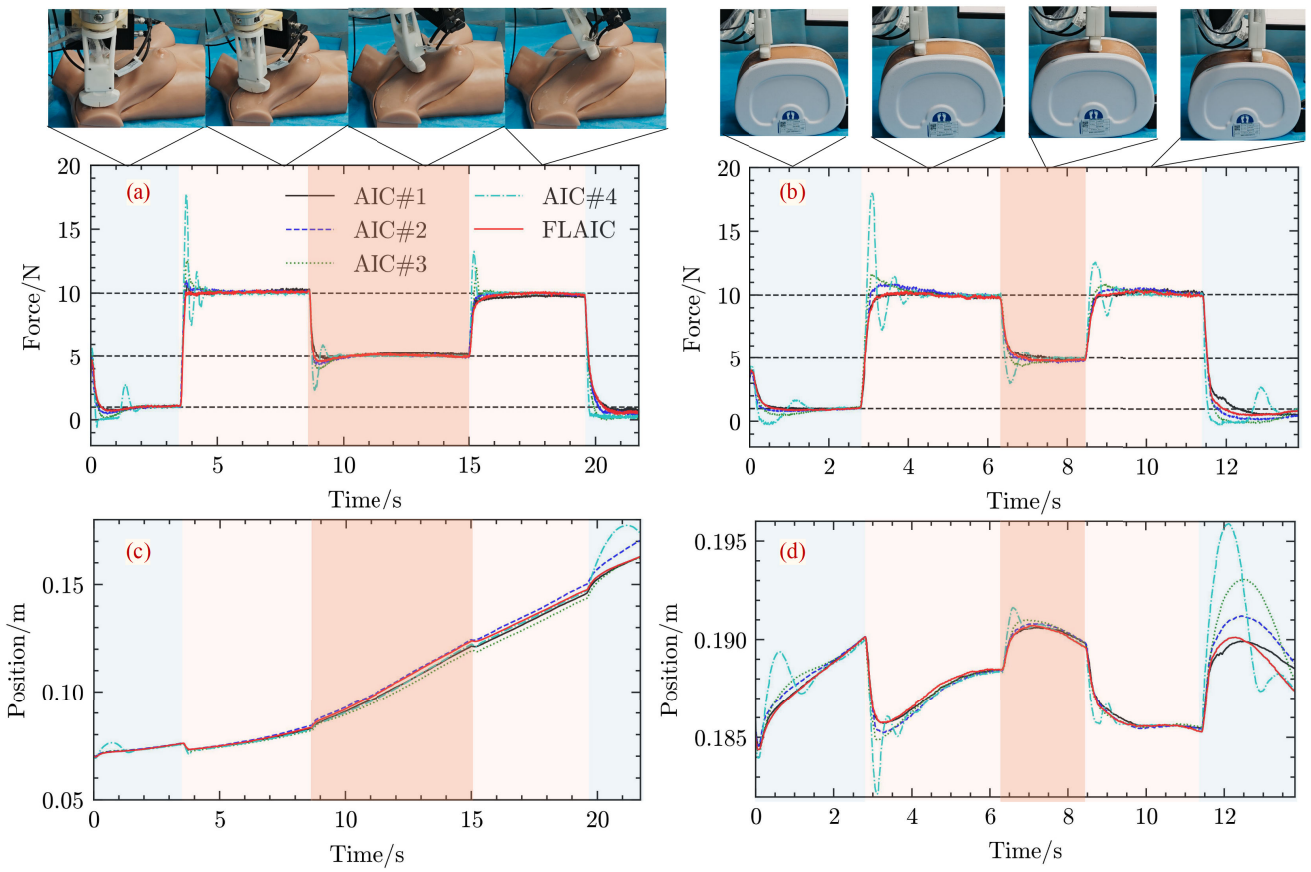


Fig. 8. Validation of FLAIC controller on phantoms. (a) Robotic breast US scanning. (b) Robotic abdomen US scanning.

TABLE I
PHANTOM EXPERIMENTS IN REAL-WORLD RUSS VALIDATIONS

Control policy	Abdomen Phantom			Breast Phantom		
	Settling time (s)	Overshoot (Worst-case) (N)	Force error (N)	Settling time (s)	Overshoot (Worst-case) (N)	Force error (N)
AIC#1	0.42± 0.23	**0.93 ± 1.74 (4.17 ± 1.05)**	0.47 ± 0.01	0.40 ± 0.21	1.16 ± 1.81 (**3.84 ± 0.41**)	0.78 ± 0.73
AIC#2	0.46 ± 0.43	1.40 ± 1.58 (4.12 ± 1.27)	0.55± 0.01	0.69 ± 0.49	1.47 ± 1.49 (4.31 ± 0.16)	0.82± 0.79
AIC#3	0.98 ± 0.45	1.91 ± 1.67 (4.62 ± 1.32)	0.55± 0.04	0.84 ± 0.51	2.05 ± 1.39 (4.51 ± 0.73)	0.84± 0.76
AIC#4	1.28± 0.53	4.01± 2.57 (8.34 ± 0.51)	0.74 ± 0.07	0.92± 0.40	3.90± 2.05 (6.97 ± 0.86)	0.90 ± 0.82
FLAIC	**0.36 ± 0.15**	1.01 ± 1.80 (4.31 ± 1.41)	**0.44 ± 0.01**	**0.33 ± 0.15**	**0.90 ± 1.54** (4.18 ± 0.23)	**0.74 ± 0.74**

fails to track rapidly changing force profiles, whereas the AIC methods exhibit significant performance diversity. Owing to the inherent sensitivity of AIC to its hyper-parameters, achieving consistently optimal behavior in practice is not easy, since AIC#1 and AIC#1 are difficulty to track efficiently to the target forces while AIC#4 demonstrates large force overshoots when target forces changed. In contrast, the proposed FLAIC consistently outperforms all baseline methods, demonstrating superior robustness and tracking accuracy under the aggressive force commands and will obtain superior fault tolerance in clinical deployment.

D. Real-World Robotic Validation

The learned weights were transferred to the developed RUSS platform shown in Fig. 5. To rigorously evaluate the proposed FLAIC, we compared the variable-force tracking accuracy of FLAIC with the same four AIC variants in CoppeliaSim simulation in both phantom and in-vivo experiments. IC was excluded from the comparison because its poor tracking accuracy raises safety concerns in clinical scenarios. Guided by clinical practice, we set $F_{\text{Non}} = 10\,\text{N}$ to guarantee adequate probe–skin contact in non-symptomatic regions, and $F_{\text{Sym}} = 5\,\text{N}$ in symptomatic regions to balance patient comfort with US image quality. The robotic experiments are recorded at https://www.youtube.com/playlist?list=PLIwEsT5216s-57PC2UUVi_tY3aLCOC2q.

(1) Phantom study: To validate the safety and tracking performance of the learned FLAIC policy, we conducted robotic variable force tracking experiments using a breast phantom (Phantom S230.52, Gaumard Inc., USA) and an abdomen phantom (Model 057A, CIRS Inc., USA), which are made of silicone material and feature complex contact surfaces. For comparison, we also evaluate the AIC algorithm with different hyperparameter along the same tracking trajectory. Fig. 8 shows the variable force tracking curves and

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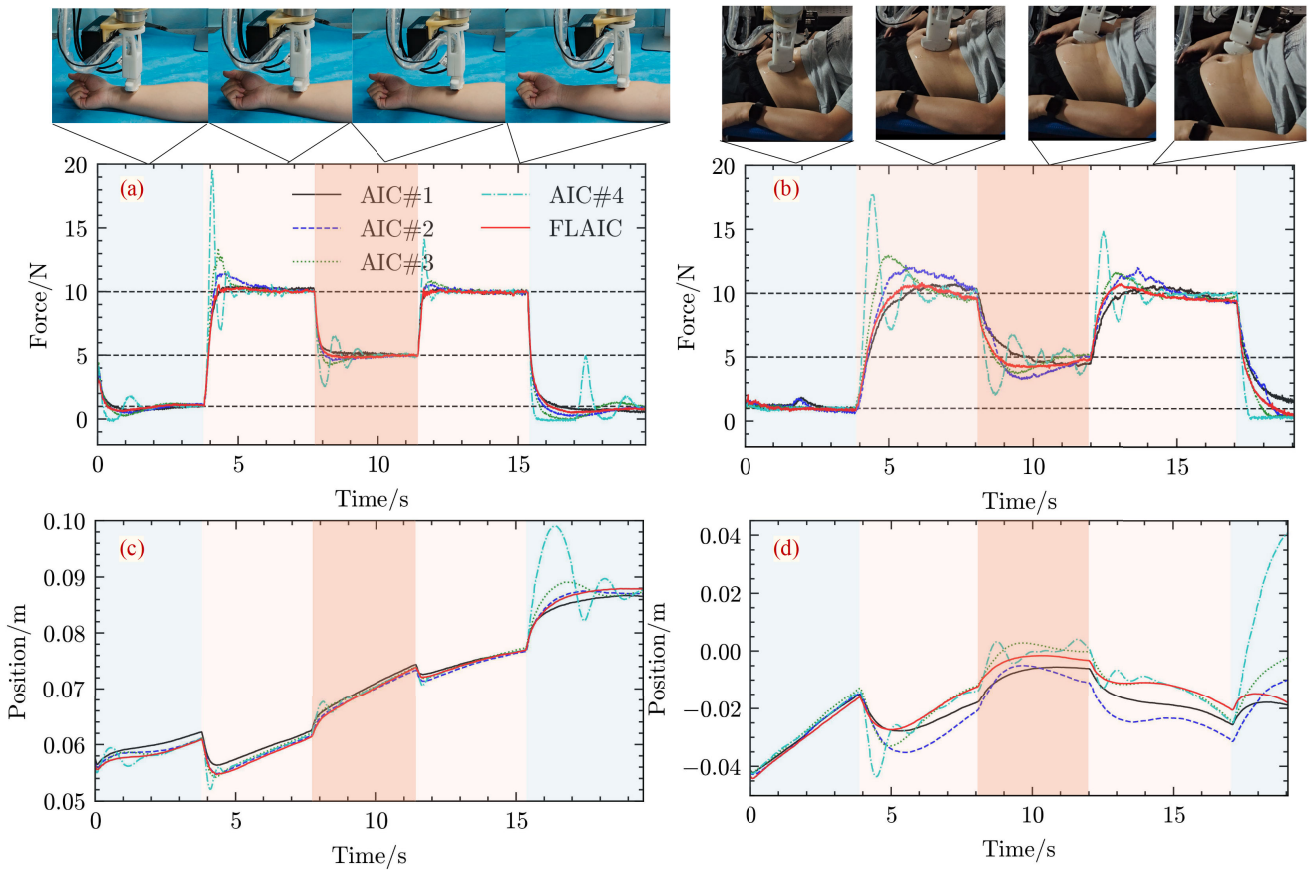


Fig. 9. Validation of LAIC controller on human-beings.

TABLE II
IN VIVO EXPERIMENTS VIA THE DEVELOPED RUSS SYSTEM

Control policy	Upper limb			Abdomen		
	Settling time (s)	Overshoot (Worst-case) (N)	Force error (N)	Settling time (s)	Overshoot (Worst-case) (N)	Force error (N)
AIC#1	0.37± 0.10	**0.68 ± 0.99** (2.30 ± 0.89)	0.59 ± 0.30	1.25± 1.11	**0.30 ± 0.36 (0.44 ± 0.24)**	1.05 ± 0.10
AIC#2	0.84 ± 0.87	1.26± 1.11 (2.85 ± 1.50)	0.60± 0.17	1.19 ± 1.09	1.23± 1.20 (3.24 ± 0.84)	1.14± 0.15
AIC#3	1.27 ± 0.68	1.85± 1.38 (3.87 ± 1.58)	0.57± 0.10	1.63 ± 0.97	1.94± 1.38 (4.32 ± 0.61)	1.02± 0.19
AIC#4	1.50± 1.20	4.03 ± 3.56 (10.57 ± 0.57)	0.63 ± 0.04	1.58± 1.11	3.76 ± 3.42 (9.62 ± 1.85)	1.26 ± 0.57
FLAIC	**0.35 ± 0.14**	0.79 ± 0.97 (**2.19 ± 1.29**)	**0.50 ± 0.19**	**0.96 ± 0.61**	0.57 ± 0.46 (1.05 ± 0.58)	**0.93 ± 0.18**

the corresponding US probe position in Z-axis of different methods under the same scanning trajectory, highlighting that the proposed FLAIC achieves lower overshoot and more stable tracking accuracy. Table I demonstrates the force tracking performance in terms of settling time, overshoot (including worst-case overshoot), and mean force error with three different trajectories. The results indicate that FLAIC significantly outperforms the AIC methods.

(2) In vivo study: Upper limb deep vein US scanning is an US-based procedure designed to evaluate the patency of the upper limb deep veins and detect the presence of thrombi, stenosis, or other pathological conditions [40]. Motivated by the superior performance of FLAIC in phantom studies, we have extended its application to in vivo upper limb deep vein US scanning. Furthermore, we conducted a comparative analysis of four AIC methods (i.e., AIC#1, AIC#2, AIC#3 and AIC#4) under this procedure. The statistical results from three different scanning trajectories are presented in Table II. As indicated in Table II, the proposed FLAIC method exhibits favorable clinical applicability, characterized by minimal overshoot, which ensures the safety of the US scanning process, and high force tracking accuracy, as evidenced by the settling time and mean force error metrics. Notably, the hyperparameters of AIC are challenging to set for different patients due to variations in skin stiffness and shape. In contrast, FLAIC demonstrates superior tracking capability and adaptability without the need for tedious manual hyperparameter tuning. Fig. 9(a) and Fig. 9(c) depict the force tracking trajectories and position variations of different force tracking methods in the upper limb deep vein US scanning procedure. From these two figures, we can found that the proposed FLAIC method achieves high tracking accuracy to the target force with minimal deviation and also smaller overshoots in upper limb RUSS scanning, thereby fulfilling clinical requirements and ensuring patient comfort in a superior manner.

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TABLE III
ABLATION STUDY OF FLAIC ON THE BREAST MODEL

Control policy	Settling time (s)	Overshoot (Worst-case) (N)	Force error (N)
FLAIC	0.33 ± 0.15	0.90 ± 1.54 (4.18 ± 0.23)	0.74 ± 0.74
-w/o. Fuzzy	0.72 ± 0.42	1.87 ± 1.42 (4.32 ± 0.57)	0.76 ± 0.71
-w/o. Enhan.	0.63 ± 0.43	1.43 ± 1.53 (4.24 ± 0.37)	0.82 ± 0.79
-w/o. ℓ_2-norm	0.42 ± 0.21	1.16 ± 1.81 (4.04 ± 0.43)	0.78 ± 0.69

TABLE IV
COMPARISON AMONG FLAIC AND OTHERC OFFLINE LEARNING-BASED AIC METHODS

Control policy	Settling time (s)	Overshoot (Worst-case) (N)	Force error (N)
FLAIC	0.33 ± 0.15	0.90 ± 1.54 (4.18 ± 0.23)	0.74 ± 0.74
LAIC [32]	0.39 ± 0.23	1.13 ± 1.79 (4.25 ± 0.45)	0.75 ± 0.69
ORL#1 [18]	0.74 ± 0.47	1.82 ± 1.43 (4.37 ± 0.51)	0.83± 0.86
ORL#2 [18]	0.75 ± 0.49	1.45 ± 1.49 (4.31 ± 0.16)	0.78± 0.81
BC [41]	0.87± 0.51	2.97± 2.15 (5.21 ± 0.79)	0.89 ± 0.82

Abdominal US scanning is also widely used in various clinical applications, such as liver ultrasound scanning and kidney ultrasound imaging. During in vivo experiments, the significant abdominal movements caused by patient respiration can severely affect the precision of robotic variable force tracking. FLAIC and four AIC methods are evaluated in this dynamic RUSS scenario, and the results are shown in Fig. 9(b) and Fig. 9(d). From Fig. 9(b), it can be observed that the proposed FLAIC achieves better tracking precision with smaller overshoot in dynamic RUSS scenarios caused by respiration. Moreover, Fig. 9(d) indicates that some AIC methods exhibit considerable pose fluctuations, which may cause patient anxiety during the robotic scanning process. In contrast, the proposed FLAIC demonstrates smoother pose variations, enhancing its clinical applicability. Additionally, the above force tracking methods were compared in terms of settling time, overshoot (including worst-case overshoot), and mean force error, as presented in Table II. The results in Table II further confirm the superior performance of the proposed FLAIC method. The above in-vivo experiments were conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board (Approval No. KY2024-025). All participants provided written informed consent prior to the study.

E. Ablation Study

To quantify the contribution of each component in FLAIC, we conduct experiments on the breast phantom (S230.52, Gaumard Inc., USA). All methods use the same offline dataset, low-level force controller, state inputs, and discrete adaptation-rate action set. We test three variants: FLAIC w/o Fuzzy, which replaces the proposed fuzzy learning representation with an original kernel feature mapping [18]; FLAIC w/o Enhancement nodes, which disables the enhancement layer; and FLAIC w/o ℓ_2-norm, which removes ridge regularization in least-squares policy evaluation by setting $\beta = 0$. Table III shows that each component contributes to performance. The fuzzy module is the most critical: removing it increases the settling time from 0.33 s to 0.72 s and the mean overshoot from 0.90 N to 1.87 N. Disabling enhancement nodes further slows the response, and removing ℓ_2 regularization slightly worsens overshoot, indicating reduced numerical robustness. Overall, the full FLAIC achieves the best force tracking behavior with low overshoot.

We further compare FLAIC with representative offline learning baselines trained on the same dataset. The compared methods include LAIC [32], which uses BLS to approximate the critic of ORL, and two offline RL baselines based on [18]. Specifically, ORL#1 uses a third-order polynomial feature mapping, and ORL#2 uses a kernel-based feature mapping; the feature formulations follow the standard settings reported in [18]. In addition, we implement a supervised imitation learning baseline, BC (Behavior Cloning) [41], which directly maps the state to a discrete adaptation-rate action using a multilayer perceptron with two hidden layers of 256 units each. As shown in Table IV, FLAIC achieves the fastest settling (0.33 s) and the lowest mean overshoot (0.90 N). These results support that the proposed fuzzy learning representation improves robustness for contact-rich variable force tracking, facilitating reliable robotic US scanning.

V. CONCLUSION

In unstructured contact-rich tasks, robotic variable force tracking faces significant challenges, such as poor tracking accuracy and substantial overshoot, particularly in scenarios like RUSS where patients have symptomatic areas, which can compromise patient safety. To address these challenges, this paper proposes a novel ℓ_2-FBORL approach for robot learning. ℓ_2-FBORL is the first FBLS-based ORL method that fully leverages the strengths of fuzzy systems and BLS, enabling the learning of near-optimal strategies from limited expert demonstration data. Based on ℓ_2-FBORL, the conventional AIC control algorithm in variable force tracking problems is modeled as an MDP, leading to the development of the FLAIC algorithm, which is the first learning-based variable force tracking algorithm in fuzzy learning paradigm. Theoretical analysis demonstrates that ℓ_2-FBORL and its application algorithm FLAIC possess universal approximation capability and convergence performance. Extensive experiments, including simulation, phantom, and robotic in vivo experiments, validate the clinical applicability and multi-scenario stability of FLAIC, demonstrating the clinical potential of the proposed FLAIC method in RUSS scenarios.

Despite the encouraging experimental results, several limitations should be acknowledged, many of which are inherent to ORL in safety-critical robotic applications. First, the proposed FLAIC framework focuses on regulating the normal contact force along a single axis. While this formulation is sufficient for a wide range of robotic ultrasound scanning tasks, it does not explicitly consider multi-axis force/torque coupling or probe orientation adaptation. Extending the proposed offline learning framework to full six-degree-of-freedom interaction control constitutes an important direction for future research. Second, consistent with common practice in ORL for physical systems, the training dataset is collected from a limited number of phantom trajectories generated by stabilizing controllers under strict clinical safety constraints. Such restricted data coverage may limit policy generalization under distribution shift [17], a well-known challenge in ORL. Incorporating larger and more diverse datasets, including multi-patient and multi-anatomy trajectories, as well as context-aware or

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distribution-robust learning strategies [22], is expected to further enhance robustness. Finally, this work primarily compares FLAIC with conventional impedance-based control methods that are widely adopted in clinical robotic systems. Although recent deep offline reinforcement learning algorithms, such as CQL [42] and TD3 +BC [43], have demonstrated strong performance in benchmark domains, deploying and evaluating such methods on patient-facing robotic platforms remains challenging due to safety, interpretability, and system integration considerations. Systematic comparisons in high-fidelity simulation or controlled bench-top environments therefore remain an important topic for future investigation.

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